

(a) At index 0, y is one-hot with 1.

$$-\sum_w y_w \log \hat{y}_w = -1 \cdot \log \hat{y}_0 = -\log \hat{y}_0$$

(b) (i) $\hat{y} = \text{softmax}(U^T v_c)$,

$$J = -\log \hat{y}_0,$$

$$J = -u_0^T v_c + \log \left(\sum_w \exp(u_w^T v_c) \right);$$

$$\frac{\partial J}{\partial v_c} = -u_0 + \sum_w \frac{\exp(u_w^T v_c)}{\sum_{w'} \exp(u_{w'}^T v_c)} u_w$$

$$= -u_0 + \sum_w \hat{y}_w u_w$$

$$= U \hat{y} - u_0$$

$$= U(\hat{y} - y)$$

(ii) $U(\hat{y} - y) = 0$, when $\hat{y} = y$, the model predicts the true outside word with probability 1.

(iii) Subtracting the gradient moves v_c towards u_0 (positive term $-u_0$), and moves away from other u_w in proportion to their predicted probabilities $\hat{y}_w$.

(c) If vector norm encodes useful magnitude like strength or frequency,

for example, "good" and "excellent" may have

the same direction, but excellent has stronger meaning with a larger magnitude, so if the tasks rely on the magnitude,

e.g. intensity-sensitive sentiment classification then the L2-normalization might remove that.

But if the task is only rely on direction, e.g. cosine similarity, then the normalization is no harm to the task.

$$(d). J = -\log \hat{y}_0 \quad \text{and} \quad \hat{y}_w = \text{softmax}(U^T v_c),$$

$\Rightarrow$ for each vocabulary word w ,

$$\begin{aligned} \frac{\partial J}{\partial u_w} &= \sum_k \frac{\partial J}{\partial \hat{y}_k} \frac{\partial \hat{y}_k}{\partial (u_w^T v_c)} \frac{\partial (u_w^T v_c)}{\partial u_w} \\ &= \left(-\frac{y_0}{\hat{y}_0} \delta_{k,0}\right) \cdot (\hat{y}_k (\delta_{k,w} - \hat{y}_w)) \cdot v_c \\ &= (\hat{y}_w - y_w) \cdot v_c \end{aligned}$$

$$w=0 \Rightarrow (\hat{y}_0 - 1) v_c$$

$$w \neq 0 \Rightarrow \hat{y}_w v_c$$

$$(e) \frac{\partial J}{\partial u} = \mathcal{V}_c (\hat{y} - y)^T$$

$$(f) f(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0 \end{cases}$$

$$(g) \sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \left(1 - \frac{1}{1 + e^{-x}}\right) = \sigma(x)(1 - \sigma(x))$$

$$\begin{aligned} (h) (i) \quad \frac{\partial J_{\text{neg}}}{\partial \mathcal{V}_c} &= -\frac{1}{\sigma} \cdot \sigma'(\mathcal{U}_0^T \mathcal{V}_c) \cdot \mathcal{U}_0 - \sum_{s=1}^k \frac{1}{\sigma(\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c)} \cdot \sigma'(-\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c) \cdot (-\mathcal{U} \mathcal{W}_s) \\ &= -(1 - \sigma(\mathcal{U}_0^T \mathcal{V}_c)) \mathcal{U}_0 + \sum_{s=1}^k (1 - \sigma(-\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c)) \cdot \mathcal{U} \mathcal{W}_s \\ &= (\sigma(\mathcal{U}_0^T \mathcal{V}_c) - 1) \mathcal{U}_0 + \sum_{s=1}^k \sigma(\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c) \mathcal{U} \mathcal{W}_s \end{aligned}$$

$$\begin{aligned} \frac{\partial J_{\text{neg}}}{\partial \mathcal{U}_0} &= -\frac{1}{\sigma(\mathcal{U}_0^T \mathcal{V}_c)} \cdot \sigma'(\mathcal{U}_0^T \mathcal{V}_c) \cdot \mathcal{V}_c \\ &= (\sigma(\mathcal{U}_0^T \mathcal{V}_c) - 1) \cdot \mathcal{V}_c \end{aligned}$$

$$\begin{aligned} \frac{\partial J_{\text{neg}}}{\partial \mathcal{U} \mathcal{W}_s} &= -\frac{1}{\sigma(-\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c)} \cdot \sigma'(-\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c) \cdot (-\mathcal{V}_c) \\ &= \sigma(\mathcal{U} \mathcal{W}_s^T \mathcal{V}_c) \mathcal{V}_c. \end{aligned}$$

(ii) Let $\mathcal{U}_0, \{\mathcal{W}_s\} = [\mathcal{U}_0, -\mathcal{U} \mathcal{W}_1, \dots, -\mathcal{U} \mathcal{W}_k]$
and define $g = \sigma(\mathcal{U}_0^T, \{\mathcal{W}_s\} \mathcal{V}_c) - 1$

$$\Rightarrow \frac{\partial J_{\text{neg}}}{\partial \mathcal{V}_c} = \mathcal{U}_0, \{\mathcal{W}_s\} g$$

(iii) Because it computes gradients only for the true outside word and a small set of K negative words (total $K+1$ vectors) instead of summing over the entire vocabulary.

$$(i) \quad \frac{\partial J_{\text{neg}}}{\partial u w_s} = - \frac{1}{\sigma(-u w_s^T v_c)} \cdot \sigma'(-u w_s^T v_c) \cdot (-v_c) \\ = \sigma(u w_s^T v_c) \cdot v_c$$

$$\frac{\partial J_{\text{neg}}}{\partial u w_s} = \left(\sum_{\substack{i=1 \\ w_i = w_s}}^K 1 \right) \sigma(u w_s^T v_c) \cdot v_c$$

$$(j) \quad \frac{\partial J_{\text{skip-gram}}}{\partial u} = \sum_{j \neq 0} \frac{\partial J(v_c, w_{t+j}, u)}{\partial u}$$

$$\frac{\partial J_{\text{skip-gram}}}{\partial v_c} = \sum_{j \neq 0} \frac{\partial J(v_c, w_{t+j}, u)}{\partial v_c}$$